

SLO-06: Site Reliability Guardians

Series: SLO — Service Level Objectives | **Notebook:** 6 of 6 | **Created:** August 2026
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Overview

An SLO answers *"is this service reliable over time?"* — evaluated continuously against a rolling or calendar window (SLO-01). A **Site Reliability Guardian (SRG)** answers a narrower, sharper question: *"is this specific change safe to release?"* — a set of DQL-based objectives validated once, at a point in time, against a deployment or business event. This notebook covers what a guardian validates, how objectives and thresholds work, creating and triggering guardians, provisioning them as code, where guardians plug into a CI/CD pipeline, and how guardians and SLOs complement rather than compete with each other.

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Prerequisites

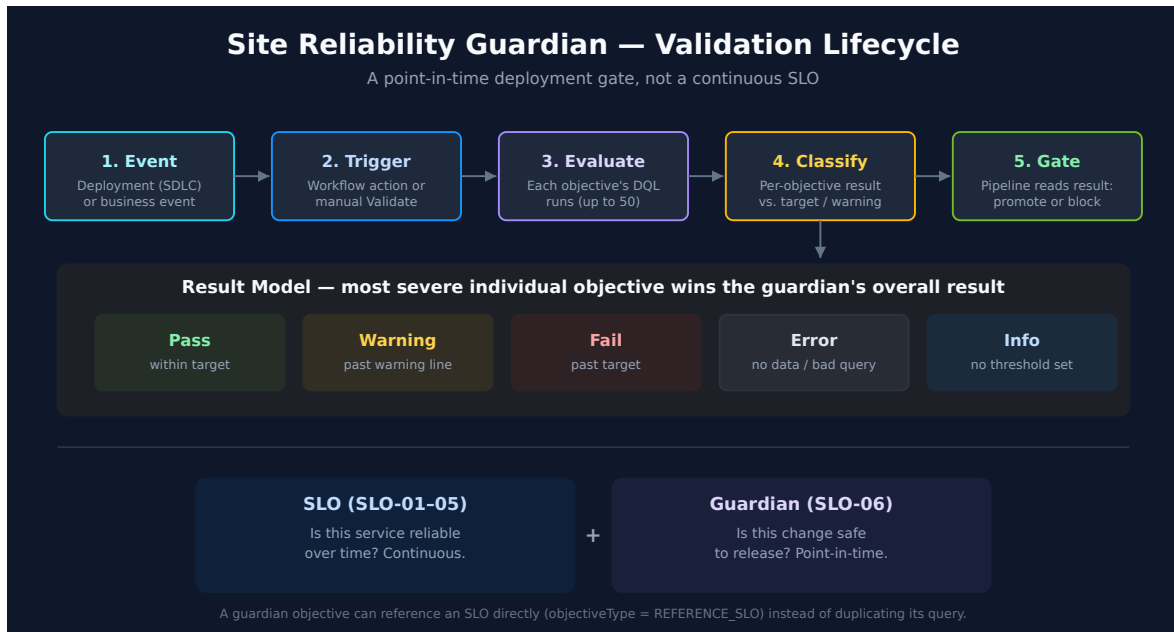
Requirement	Details
Dynatrace Environment	SaaS Gen3 with the Site Reliability Guardian app
Permissions	Grail read scopes covering whatever data your objective DQL queries touch (<code>storage:spans:read</code> , <code>storage:metrics:read</code> , <code>storage:bizevents:read</code> , ...); workflow authoring permissions for automated validation
Tooling (as code)	Terraform with the <code>dynatrace-oss/dynatrace</code> provider, or Monaco
Prior reading	SLO-01 (SLI/SLO fundamentals), SLO-02 (writing DQL SLI queries — the same DQL skill authors guardian objectives)

1. What a Guardian Validates

A **Site Reliability Guardian** is a Dynatrace app that automates change impact analysis to validate service availability, performance, and capacity objectives for a set of guarded entities. Where an SLO is a standing promise evaluated continuously, a guardian is a **point-in-time validation** run against a specific event. The app supports two guardian types:

Guardian type	eventKind	Triggered by
Lifecycle guardian	<code>SDLC_EVENT</code>	Software delivery events — a deployment, a release-stage transition
Business guardian	<code>BIZ_EVENT</code>	Business events — a marketing-campaign launch, a pricing change

Both guardian types share the same mechanics: a named group of **objectives**, each one a DQL query evaluated against a chosen timeframe, checked against thresholds, and rolled up into a single result for the guardian as a whole.



2. Objectives, Thresholds, and the Result Model

Each objective is, at its core, a DQL query returning a single scalar value, checked against a threshold. This is the same DQL skill from SLO-02 — a guardian objective and an SLO's SLI query are structurally the same *good ÷ total* or scalar-metric shape. What differs is the timeframe (a short deployment window, not a rolling 30 days) and what happens with the result (gate a release, not accumulate an error budget).

Objective type

objectiveType	What it evaluates
DQL	A DQL query you write, evaluated over the guardian's validation timeframe
REFERENCE_SLO	An existing SLO's current value — ties a guardian directly to an SLO defined in SLO-01/02 rather than duplicating the query

Threshold type

Threshold type	Behavior
No thresholds	Informational only — the objective reports a value but never fails the guardian

Threshold type	Behavior
Static thresholds	Fixed <code>target</code> (fail point) and <code>warning</code> (soft threshold) values
Auto-adaptive thresholds	Threshold adjusts dynamically from historical data — the same idea as an SLO's adaptive baseline

A `comparisonOperator` says which direction is good — "a higher value is good" (`GREATER_THAN_OR_EQUAL`) or "a lower value is good" (`LESS_THAN_OR_EQUAL`). An error-rate objective is a lower-is-good comparison against a `target` / `warning` pair; a request-count or bizevent-count objective is usually higher-is-good.

Result model

Each objective resolves to one of five states, and **the guardian's overall result is the most severe individual objective result**:

Result	Meaning
Pass	Value within the acceptable range
Warning	Value crossed the warning threshold but not the fail threshold
Fail	Value crossed the fail threshold
Error	The indicator could not be derived (bad query, no data)
Info	Informational objective (no threshold) — never drives Fail

Two objective queries below, live-verified against the validation tenant, in the shape a guardian would actually run them — a short window, since a guardian validates one change, not a rolling reliability posture.

Error rate — lower is good (`LESS_THAN_OR_EQUAL` comparison):

```
// Guardian objective – error rate. Lower is good; compare against
target/warning with LESS_THAN_OR_EQUAL.
// Live-verified against the validation tenant: returned ~0.40% over the last
hour.
fetch spans, from:-1h
| filter span.kind == "server"
| summarize {
    good = countIf(duration < 500ms),
    total = count()
}
| fieldsAdd errorRate = 100.0 - (good * 100.0) / total
| fields errorRate, good, total
```

Availability — higher is good (GREATER_THAN_OR_EQUAL comparison):

```
// Guardian objective – availability. Higher is good; compare against
target/warning with GREATER_THAN_OR_EQUAL.
// Live-verified against the validation tenant: returned ~95.6% over the last
30 minutes.
timeseries {
    total = sum(dt.service.request.count),
    failures = sum(dt.service.request.failure_count)
}, from:-30m
| fieldsAdd availability = ((arraySum(total) - arraySum(failures)) /
arraySum(total)) * 100
| fields availability
```

3. Creating a Guardian

Two ways to create a guardian in the app:

- **From a template** — pick a template, run its prefilled DQL query to select guarded entities (or select all rows on the current page), then refine the pre-populated objectives.
- **Blank** — set the guardian type and name, then add objectives with your own DQL queries. When you select an entity, the app prefills the query with the correct scoping — replace any remaining placeholder (e.g. `HOST-PLACEHOLDER`) with your actual value.

Limit: a guardian supports a maximum of **50 objectives**.

Variables parameterize a guardian at the group level — define a guardian-level variable (prefixed `$`, e.g. `$MyTotal`), reference it inside an objective's DQL query, and set its

value either at creation time or per validation run. This is what makes one guardian definition reusable across environments (dev / staging / prod) instead of hardcoding a threshold or entity per guardian copy.

4. Triggering Validation

A guardian's objectives don't evaluate themselves — validation is triggered one of two ways:

- **On-demand** — an operator opens the guardian and selects **Validate**. Useful while authoring objectives, or for an ad hoc check.
- **Automated, via Workflow** — a workflow with an event trigger runs the `dynatrace.site.reliability.guardian:validate-guardian-action` task, which validates a named guardian over a timeframe (e.g. the last 30 minutes) whenever a matching event arrives. A representative trigger filter: `application == "easytrade" and stage == "production"` — scoping validation to the specific deployment that just happened, not every deployment across every service.

This is ordinary Workflow mechanics — event trigger, filter condition, action — covered in general by the **WFLOW series**. What's specific to guardians is the action type and the fact that its result (Pass / Warning / Fail) is what a CI/CD pipeline reads back to decide whether to proceed (§6).

5. Guardians as Code

A guardian evaluated only by hand in the UI doesn't scale past one team's first pipeline. Provision it the same way SLO-05 provisions SLOs.

Terraform

The `dynatrace_site_reliability_guardian` resource — schema confirmed against the `dynatrace-oss/dynatrace` provider registry, 08/31/2026:

```
# dynatrace_site_reliability_guardian – schema confirmed against the
# dynatrace-oss/dynatrace provider registry, 08/31/2026.
resource "dynatrace_site_reliability_guardian" "checkout_release_gate" {
  name          = "Checkout Release Gate"
  event_kind    = "SDLC_EVENT"
  tags          = ["stage:staging"]

  objectives {
    objective {
      name              = "Error rate"
      objective_type    = "DQL"
      comparison_operator = "LESS_THAN_OR_EQUAL"
      dql_query         = <<<-EOT
        fetch spans, from:-30m
        | filter span.kind == "server"
        | summarize good = countIf(duration < 500ms), total = count()
        | fieldsAdd errorRate = 100.0 - (good * 100.0) / total
        | fields errorRate
      EOT
      target = 8
      warning = 6
    }
    objective {
      name              = "Checkout volume"
      objective_type    = "DQL"
      comparison_operator = "GREATER_THAN_OR_EQUAL"
      dql_query         = <<<-EOT
        fetch bizevents, from:-30m
        | filter event.type == "checkout"
        | summarize count()
      EOT
      target = 50
      warning = 55
    }
  }
}
```

A few things this schema encodes, and where it diverges from the SLO-05 Terraform resource so you don't carry that field naming over by mistake:

- **target** / **warning**, not SLO-05's **target_success** / **target_warning** — two different resources, two different argument names for a similar idea.
- **comparison_operator** takes exactly **GREATER_THAN_OR_EQUAL** or **LESS_THAN_OR_EQUAL** — not the SLO app's own comparison vocabulary.

- `objective_type = "REFERENCE_SLO"` is the alternative to `DQL` — set `reference_slo` to an existing SLO instead of writing a query, which is the as-code form of §7's "guardian objective backed by an SLO" pattern.
- `event_kind` is `SDLC_EVENT` or `BIZ_EVENT` — matching the guardian-type distinction from §1.

Monaco

Monaco projects use a three-file convention rather than a single settings object:

File	Holds
<code>guardian.json</code>	The guardian definition — name, <code>eventKind</code> , tags, and the <code>objectives</code> array (<code>objectiveType</code> , <code>dqlQuery</code> or a reference, <code>target</code> , <code>warning</code> , <code>comparisonOperator</code>)
<code>workflow.json</code>	The automation template that triggers validation — an event trigger with a filter condition, and a task invoking the guardian-validate action
<code>config.yaml</code>	Shared parameters both files reference, so one edit updates both the guardian and the workflow that validates it

The dtctl gap

As of the `dtctl` version this repo standardizes on (0.38 — see the root `CLAUDE.md`), `dtctl get --help` lists no dedicated `guardians` resource. Terraform and Monaco are the working config-as-code path today; there is no `dtctl get guardians` / `dtctl apply` shortcut to reach for instead.

6. CI/CD Pipeline Integration

The guardian mechanics above only pay off wired into a pipeline. The shape is consistent across CI/CD platforms:

1. **Deploy** the change to the target environment.
2. **Emit a deployment event** (an SDLC event, if using a lifecycle guardian) so Dynatrace has something to correlate the validation window against.
3. **Trigger the guardian workflow** — either directly from the pipeline (call the workflow/guardian API) or let the event trigger from step 2 fire it automatically (§4).

4. **Read back the result.** A Pass gates the pipeline forward; a Fail blocks promotion; a Warning is a judgment call — treat it like the slow-burn tier in SLO-04, a signal to the team rather than an automatic block.

AUTOM-07 documents the working samples for this: `github_pipeline_observability` / `gitlab_pipeline_observability` / `argocd_observability` (Monaco and Terraform variants) ingest SDLC events via OpenPipeline with pre-built dashboards, and the community [dynatrace-automation-tools](#) CLI wraps guardian validation and deployment-event ingestion specifically for CI/CD use — see AUTOM-07's Community Resources section for the full sample matrix across GitHub Actions, GitLab CI, Azure DevOps, and ArgoCD.

7. Guardian vs SLO — Which One, or Both

	SLO (SLO-01–05)	Site Reliability Guardian
Question answered	Is this service reliable over time?	Is this specific change safe to release?
Evaluation	Continuous, rolling or calendar window (SLO-01 §2)	Point-in-time, triggered by an event
Output	Error budget consumption, burn rate (SLO-03)	Pass / Warning / Fail / Error / Info per objective
Alerting fit	Burn-rate alerting, ongoing (SLO-04)	Pipeline gate — block or promote a deployment
As code	<code>dynatrace_platform_slo</code> (SLO-05)	<code>dynatrace_site_reliability_guardian</code> (§5)

They are not competing mechanisms — a guardian objective can reference an SLO directly (`objective_type = "REFERENCE_SLO"`, §5), which is the natural way to use both together: **the SLO defines what "reliable" means for the service on an ongoing basis; the guardian asks whether the change you're about to ship keeps it that way.** A team with no SLOs yet can still use guardians standalone with DQL objectives (§2) — a guardian doesn't require an SLO to exist first — but a mature setup typically has SLOs defining the target and guardians enforcing it at every release.

Summary

In this notebook you learned:

1. **What a guardian validates** — point-in-time change validation, not continuous reliability; Lifecycle (`SDLC_EVENT`) vs Business (`BIZ_EVENT`) guardian types
2. **Objectives and thresholds** — `DQL` vs `REFERENCE_SLO` objective types, static / adaptive / no-threshold modes, and the five-state Pass / Warning / Fail / Error / Info result model (most severe wins)
3. **Creating a guardian** — template vs blank, the 50-objective limit, guardian-level variables
4. **Triggering validation** — on-demand Validate vs the automated Workflow action
5. **Guardians as code** — the `dynatrace_site_reliability_guardian` Terraform resource, the Monaco three-file convention, and the current `dtctl` gap
6. **CI/CD integration** — the deploy → event → trigger → gate shape, and where AUTOM-07's working samples plug in
7. **Guardian vs SLO** — complementary, not competing; a guardian objective can reference an SLO directly

Series Summary

Notebook	Topic	Key Takeaway
SLO-01	Fundamentals	SLI/SLO/error-budget building blocks, modern vs classic SLO app
SLO-02	Defining SLIs	Good/total DQL for availability, latency, error-rate, and business SLIs
SLO-03	Composition & Error Budgets	Burn-rate math, composite/weighted-global SLOs, rolling vs calendar windows
SLO-04	Alerting	Burn-rate over threshold-on-SLI; fast/slow-burn multiwindow alerts
SLO-05	SLOs as Code	Terraform/Monaco/API provisioning for the modern SLO app

Notebook	Topic	Key Takeaway
SLO-06	Site Reliability Guardians	Point-in-time deployment validation gates, complementary to ongoing SLOs

Sources: [Site Reliability Guardian \(DT docs\)](#), [Create a Site Reliability Guardian \(DT docs\)](#), [Add Site Reliability Guardian objective \(DT docs\)](#), [Site Reliability Guardian as code \(DT docs\)](#), [dynatrace_site_reliability_guardian resource \(Dynatrace provider docs\)](#). Terraform schema and the objective/threshold model fetched at source 08/31/2026; the two DQL objective queries in §2 were executed against the validation tenant the same day. **Derived:** the dtctl-gap note in §5 combines this repo's pinned `dtctl` version (root `CLAUDE.md`) with a direct `dtctl get --help check` — neither source states the gap on its own.

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